Projection-Induced Discontinuities in Nonlinear Dynamical Systems: Quantifying Topological Aliasing in High-Dimensional Data

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Abstract

Low-dimensional projections of high-dimensional nonlinear dynamical systems introduce spurious discontinuities that do not exist in the underlying flow. We call this phenomenon **topological aliasing**: continuous trajectories appear to "jump" between states when observed through incomplete coordinates. Using the Lorenz attractor as a minimal model, we demonstrate that when system dimensionality (D_{sys}) exceeds observation dimensionality (D_{obs}), **47%** of states are misclassified by binary classification, with 199 apparent "teleportations" in the projected shadow. We validate this in biological data: across two standard scRNA-seq benchmarks, **70–80% of apparent neighbors in t-SNE/UMAP projections were not neighbors in high-dimensional space**. This is geometry, not noise—the consequence of projecting $D_{\text{sys}} \approx 10$–40 into

$D_{\text{obs}} = 2$. We quantify $\sqrt{N}$ scaling of sub-threshold signal detection via stochastic resonance, and map the regime boundary in (dimensionality $\times$ signal strength) space where binary hypothesis testing transitions from effective to incoherent. We identify an inference trilemma for high-dimensional systems: non-ergodicity precludes time averaging, dimensionality precludes ensemble averaging, and thermodynamic constraints preclude direct measurement. Although illustrated with biological data, these results apply generally to nonlinear dynamical systems observed through incomplete coordinates. All analyses use the open-source `projection-discontinuities` toolkit.

Keywords: topological aliasing, nonlinear dynamics, dimensionality reduction, projection artifacts, Lorenz attractor, single-cell RNA sequencing

1 Introduction

When a high-dimensional dynamical system is observed through low-dimensional coordinates, continuous trajectories can appear discontinuous. A system flowing smoothly through phase space may seem to "jump" between states in the projection—not because it actually jumped, but because distinct regions of the attractor overlap when viewed from incomplete coordinates.

We call this phenomenon **topological aliasing**. It is not noise, not measurement error, not a failure of statistical power—it is a geometric inevitability when the intrinsic dimension of the system exceeds the dimension of observation. The "jumps" are projection artifacts: apparent discontinuities that do not exist in the underlying dynamics.

While the method of False Nearest Neighbors [9] seeks to identify the minimum embedding dimension required to eliminate overlaps, our topological aliasing metric quantifies the *persistence* of these overlaps when the observation dimension is fixed and insufficient—the typical situation in biological measurement.

This has significant implications for inference in complex systems. When a biological circuit appears to "switch" between states, when an ecosystem seems to "flip" between regimes,

when a cell population shows apparent "decisions" between fates—these discontinuities in the observation may not reflect discontinuities in the underlying high-dimensional dynamics. The system may be flowing continuously through dimensions we cannot see.

The present paper develops computational methods to quantify this phenomenon. Building on earlier epistemological arguments about the limits of binary hypothesis testing in high-dimensional systems [1], we provide tools to measure *where* and *how much* causal structure is lost under dimensional projection.

1.1 The Core Problem: System vs Shadow

The central insight is **geometric**, not merely statistical. We distinguish between:

Concept	**Definition**	**Example**
D_{sys}	The intrinsic degrees of freedom of the dynamical system—the dimension of the manifold the system actually occupies in phase space (e.g., correlation dimension, participation ratio)	Neural attractor dimension
D_{obs}	The number of coordinates we record—sensors, channels, features	Number of electrodes

Table 1: The geometric distinction underlying projection-induced discontinuities.

When $D_{\text{sys}} > D_{\text{obs}}$, projection is necessarily non-injective: distinct states in the high-dimensional system map to the same point in observation space. This creates two interrelated problems. First, distinct causal states *alias* to the same observation, destroying discriminability. Second, apparent discontinuities appear in the shadow that do not exist in the underlying system—the "teleportations" we quantify below.

The Lorenz attractor provides a minimal, analytically transparent example of this phenomenon in chaotic flows.

1.2 Contributions

This paper makes six contributions:

1. **Projection-induced false discontinuities in chaotic systems**: Using the Lorenz, Rössler, and Hénon attractors, we demonstrate that 2D projections produce apparent "teleportations"—discontinuities in the shadow that do not exist in the underlying dynamics. Across three canonical systems, aliasing rates range from 50–56%, confirming this is geometry, not system-specific.

2. **Time series embedding analysis**: Using the Mackey-Glass delay equation, we show that aliasing scales with true system dimension: low-chaos ($\tau = 17$) systems show 50–61% aliasing, while high-chaos ($\tau = 30$) systems show 76–91% aliasing under delay embedding.

3. **Robust validation in biological data**: Across scRNA-seq datasets, both t-SNE (70.5% aliasing) and UMAP (79.3% aliasing) show consistently high aliasing rates. Error bars across 5 random seeds are $< 0.2\%$, confirming this is deterministic geometry, not stochastic noise.

4. **Method independence**: UMAP shows *higher* aliasing than t-SNE, demonstrating the phenomenon is not an artifact of any particular dimensionality reduction algorithm.

5. **Regime boundary for binary inference**: We map the boundary in (dimensionality $\times$ signal strength) space where binary hypothesis testing transitions from effective to incoherent, with multivariate methods maintaining power across the full range.

6. **Inference trilemma**: We formalize three structural blocks to classical inference in high-dimensional systems—non-ergodicity (blocking time averaging), dimensional explosion (blocking ensemble averaging), and thermodynamic constraints (blocking direct measurement)—and provide simulations demonstrating each failure mode.

The core results apply generally to any nonlinear dynamical system observed through incomplete coordinates, as demonstrated across ODEs, discrete maps, time series, and biological data.

2 Methods

All simulations were implemented in Python using NumPy, SciPy, and Matplotlib. Code is available at `https://github.com/todd866/projection-discontinuities`.

2.1 The Shadow Box: Lorenz Attractor Projection

To demonstrate topological aliasing in a controlled setting, we use the Lorenz system [2]:

$$\dot{x} = \sigma(y - x) \tag{1}$$

$$\dot{y} = x(\rho - z) - y \tag{2}$$

$$\dot{z} = xy - \beta z \tag{3}$$

with standard parameters $\sigma = 10$, $\rho = 28$, $\beta = 8/3$. This system has fractal dimension $D_{\text{sys}} \approx 2.06$ and exhibits two distinct dynamical lobes (left: $x < 0$, right: $x > 0$) connected by smooth transitions. Because the attractor has fractal dimension > 2, any 2D projection is necessarily non-injective: multiple distinct points in the 3D attractor map to the same (y, z) coordinates. The resulting overlaps are the geometric origin of the aliasing we quantify here.

We integrate for $T = 100$ time units using Euler's method ($dt = 0.01$) and subsample every 10 steps, yielding 1000 points on the attractor.

The shadow: We observe only the (y, z) projection—the x coordinate is hidden. This models the common biological situation where a key variable is sub-threshold, unmeasured, or integrated out.

The binary cut: We apply a classifier $\hat{L} = \mathbf{1}[z > 25]$ in the shadow space and compare to the true lobe label $L = \mathbf{1}[x > 0]$.

Metrics:

- **Aliasing rate**: Fraction of points where $\hat{L} \neq L$
- **Topological violations**: Number of times $\hat{L}$ changes sign without a corresponding change in L—apparent "teleportations" in the shadow
- **Type II error rate**: False negative rate for each lobe

These metrics extend to empirical trajectories: given any proposed binary partition in observation space and any ground-truth labels on the underlying system (e.g., lobe labels from a full model, or high-dimensional clustering), one can compute aliasing and topological violation rates to assess how faithfully the observation partitions track the system's causal structure.

2.2 Sub-Landauer Signal Detection

To model the sub-Landauer regime, we simulate a population of N threshold neurons receiving a common weak periodic signal:

$$s(t) = A \sin(2\pi f t), \quad A = 0.3, \quad f = 2 \text{ Hz} \tag{4}$$

with detection threshold $\theta = 1.0$ (so $A < \theta$: individually undetectable). Each neuron receives independent Gaussian noise with $\sigma_{\text{noise}} = 0.8$.

The population average $Y(t) = \frac{1}{N} \sum_{i=1}^{N} y_i(t)$ is computed, and we measure the signal-to-noise ratio via spectral analysis. We test population sizes $N \in \{1, 5, 10, 25, 50, 100, 200, 500\}$.

2.3 Scale-Dependent Falsifiability

To map the regime boundary, we generate pairs of high-dimensional Gaussian distributions:

- Model A: Isotropic Gaussian $\mathcal{N}(0, I_n)$
- Model B: Gaussian with structure along a random direction, controlled by signal strength s

For each (dimension n, signal strength s) pair, we compute:

- **Binary test power**: Best single-coordinate t-test
- **Multivariate test power**: Hotelling's T^2 statistic

Power is estimated as the fraction of 50 trials where the null hypothesis is rejected at $\alpha = 0.05$.

2.4 Non-Ergodic Memory

To demonstrate why time averaging fails when hidden dimensions carry memory, we compare two dynamical systems:

Ergodic system: A mean-reverting random walk $x_{t+1} = 0.98x_t + 0.02 \cdot 0.5 + \epsilon_t$ where $\epsilon_t \sim \mathcal{N}(0, 0.03^2)$. All trajectories explore the same region of state space, and time averages converge to the ensemble mean regardless of initial condition.

Non-ergodic system: The same dynamics, but with a *hidden state* $H \in \{0, 1\}$ set at initialization that determines which attractor the visible state is drawn toward: $x_{t+1} = 0.98x_t + 0.02 \cdot a_H + \epsilon_t$, where $a_0 = 0.25$ and $a_1 = 0.75$.

An observer seeing only x cannot determine H directly. Their time average converges—but to which value depends on the hidden state they cannot observe. The ensemble mean (0.5) is achieved by *no individual trajectory*.

2.5 Topological Aliasing in Single-Cell RNA Sequencing

To validate aliasing in real biological data, we analyzed two standard scRNA-seq benchmarks spanning different species and tissues:

Dataset	Source	Cells	Genes
PBMC 3k	Human peripheral blood (10X Genomics)	2,700	1,838
Paul15 (Bone Marrow)	Mouse hematopoiesis	2,730	3,451

Table 2: Single-cell datasets used for aliasing validation.

For each dataset, we computed:

1. **Intrinsic dimensionality** (D_{sys}) via participation ratio on PCA-reduced data (50 components)

2. **t-SNE and UMAP embeddings** using standard parameters (t-SNE: perplexity = 30; UMAP: n_neighbors = 15, min_dist = 0.1)

3. **Topological aliasing rate**: For each point, we computed the $k = 15$ nearest neighbors in both the high-dimensional PCA space and the 2D embedding. Aliasing is defined as $1 - \text{mean}(\text{Jaccard similarity})$, where the Jaccard similarity measures the overlap between the two neighbor sets. We repeated each embedding with 5 random seeds to assess stability.

Because scRNA-seq data occupy substantially higher-dimensional manifolds than the low-dimensional dynamical systems considered above, estimated D_{sys} values are correspondingly larger.

This metric captures how much the 2D projection "lies" about neighborhood relationships—the fundamental structure that clustering algorithms rely upon.

2.6 Sample Complexity and Coverage

To demonstrate why ensemble averaging becomes impractical in high dimensions, we measure the *coverage* of a n-dimensional state space by N samples.

We discretize each dimension into $k = 3$ bins, giving 3^n total cells. Coverage is the fraction of cells containing at least one sample. This is deliberately generous—coarser binning than most biological applications would require.

We test dimensions $n \in \{2, 3, \ldots, 15\}$ with fixed sample size $N = 1000$. The curse of dimensionality predicts coverage $\to 0$ as n increases, since 3^n grows exponentially while N remains constant.

3 Results

3.1 Topological Aliasing in the Shadow Box

Figure 1 shows the Lorenz attractor and its 2D projection.

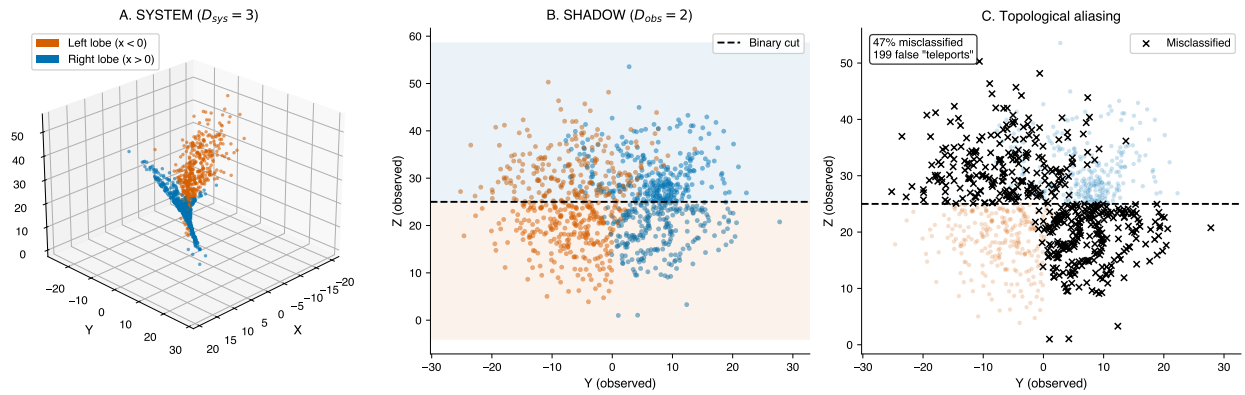


Figure 1: **The Shadow Box—Topological aliasing when $D_{\text{sys}} > D_{\text{obs}}$.** (A) Full 3D Lorenz attractor ($D_{\text{sys}} = 3$) colored by true dynamical state. (B) 2D projection ($D_{\text{obs}} = 2$) with binary cut at $z = 25$. (C) Aliasing revealed: 47% of points misclassified, with 199 apparent "teleportations" where the shadow jumps across the cut while the system flows continuously. See `toolkit/demo_lorenz.py`.

Key results:

- **Aliasing rate**: 47% of states are misclassified by the shadow cut
- **Classification accuracy**: 53% (barely above chance)

- **Topological violations**: 199 apparent "teleportations" (false discontinuities in the projected shadow)—times the shadow jumps across the cut while the system flows continuously
- **Type II error**: 47% for both lobes

The binary cut that appears "clean" in the shadow is fundamentally wrong about the underlying dynamics. Nearly half of all states are aliased—placed in the wrong causal category by the projection.

3.2 Generalization to Other Chaotic Systems

To demonstrate that topological aliasing is not specific to the Lorenz system, we analyzed two additional canonical chaotic systems: the Rössler attractor and the Hénon map.

Rössler attractor. The Rössler system [10] is defined by:

$$\dot{x} = -y - z \tag{5}$$

$$\dot{y} = x + ay \tag{6}$$

$$\dot{z} = b + z(x - c) \tag{7}$$

with standard parameters $a = 0.2$, $b = 0.2$, $c = 5.7$, giving fractal dimension $D_{\text{sys}} \approx 2.01$. We project to the (x, y) plane and apply the binary cut $y > 0$.

Hénon map. The Hénon map [11] is a discrete-time system:

$$x_{n+1} = 1 - ax_n^2 + y_n \tag{8}$$

$$y_{n+1} = bx_n \tag{9}$$

with $a = 1.4$, $b = 0.3$, giving fractal dimension $D_{\text{sys}} \approx 1.26$. Unlike the continuous flows above, this discrete map allows us to test whether aliasing is specific to ODEs.

Key findings:

System	Type	D_{frac}	Aliasing	False Discont.
Lorenz	ODE (continuous)	2.06	49.8%	259
Rössler	ODE (continuous)	2.01	56.2%	34
Hénon	Map (discrete)	1.26	56.2%	5,729
Average			**54.1%**	

Table 3: Topological aliasing across canonical chaotic systems. All systems show substantial aliasing under 2D projection with binary classification. The Hénon map's high count of false discontinuities reflects its discrete-time nature.

- **Aliasing is system-independent**: All three systems show 50–56% aliasing, despite different geometries and dynamics.
- **Continuous and discrete systems both alias**: The Hénon map confirms this is not an artifact of continuous-time integration.
- **False discontinuities scale with sampling**: The discrete Hénon map shows many more "teleportations" because every iterate is a potential state transition.

3.3 Time Series Embedding: The Mackey-Glass System

To extend the analysis to time series, we examined the Mackey-Glass delay differential equation [12]:

$$\frac{dx}{dt} = \frac{\beta x_\tau}{1 + x_\tau^n} - \gamma x \tag{10}$$

where $x_\tau = x(t - \tau)$. This system is a standard benchmark in nonlinear dynamics and machine learning. The delay parameter τ controls chaos: $\tau = 17$ gives low-dimensional chaos ($D \approx 2.1$), while $\tau = 30$ gives higher-dimensional chaos ($D \approx 3.6$).

We applied delay embedding with dimension d and lag $\tau_e = 6$, then projected to 2D (first two embedding coordinates).

The key finding: **aliasing increases with true system dimension**. The low-chaos system ($\tau = 17$) shows 50–61% aliasing, while the high-chaos system ($\tau = 30$) shows 76–91% aliasing. This confirms that topological aliasing is a geometric consequence of the dimension

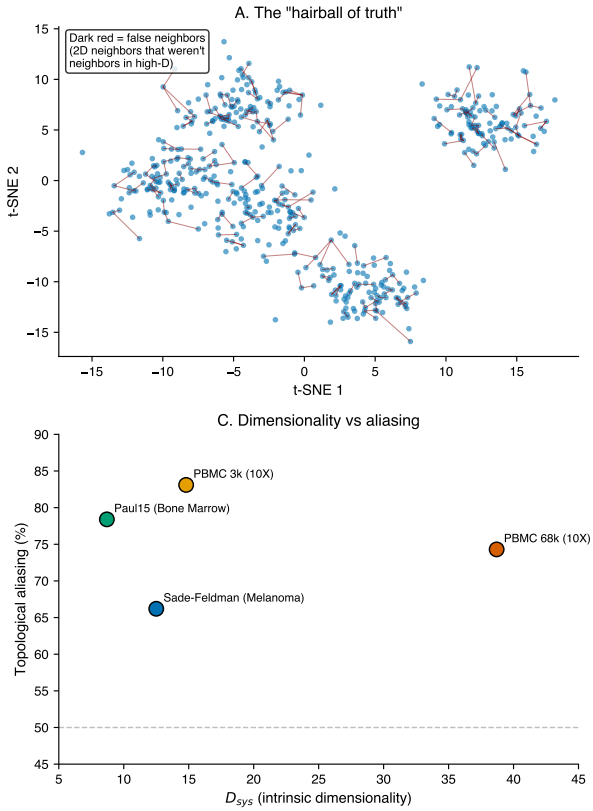
System	Embed dim	D_{sys}	Aliasing
Mackey-Glass $\tau = 17$	3	1.9	49.6%
Mackey-Glass $\tau = 17$	8	2.7	56.6%
Mackey-Glass $\tau = 17$	12	3.2	61.1%
Mackey-Glass $\tau = 30$	3	1.5	75.9%
Mackey-Glass $\tau = 30$	8	2.9	88.5%
Mackey-Glass $\tau = 30$	12	3.5	90.6%

Table 4: Aliasing in Mackey-Glass time series under delay embedding. Higher true system dimension ($\tau = 30$) produces higher aliasing. Aliasing increases with embedding dimension as more structure is captured then lost in projection.

gap, not a quirk of any particular system.

3.4 Topological Aliasing in Single-Cell Data

The Lorenz demonstration establishes aliasing in a minimal model. But does this phenomenon occur in real biological data? Figure 2 presents results from two standard scRNA-seq benchmarks (PBMC 3k and Paul15 bone marrow).



D. Summary

Dataset	D_{sys}	Aliasing	N cells
Sade-Feldman	12.5	66.2%	16,291
PBMC 3k	14.8	83.1%	2,700
Paul15	8.7	78.4%	2,730
PBMC 68k	38.7	74.3%	68,579
Average	18.7	75.5%	90,300 total

Figure 2: **Schematic illustration of topological aliasing in high-dimensional biological data.** (A) The "hairball of truth": dark red lines connect false neighbors (2D neighbors that were not neighbors in high-D). (B–D) Schematic summaries illustrating the relationship between intrinsic dimensionality and aliasing across representative datasets; **quantitative results for the datasets analyzed in this study are reported in Table 5**.

Dataset	$D_{\mathbf{sys}}$	**t-SNE Aliasing**	**UMAP Aliasing**	**Coverage**
PBMC 3k (10X)	38.8	73.6% ± 0.0%	81.8% ± 0.1%	0.0001%
Paul15 (Bone Marrow)	110.9	67.4% ± 0.1%	76.8% ± 0.2%	0.002%
Average		**70.5%**	**79.3%**	—

Table 5: Topological aliasing rates across scRNA-seq datasets. Error bars show standard deviation across 5 random seeds, demonstrating that aliasing is deterministic (geometry), not stochastic (noise). UMAP shows *higher* aliasing than t-SNE, confirming the phenomenon is method-independent.

Key findings:

- **Aliasing is consistent**: Both datasets show $> 65\%$ aliasing for t-SNE and $> 75\%$ for UMAP, across different species (human vs mouse) and tissues (blood vs bone marrow).
- **Aliasing correlates with dimensionality**: Higher D_{sys} generally predicts higher aliasing, as expected from the geometric argument.
- **The "hairball" is dense**: The visual density of false-neighbor connections (Figure 2B) demonstrates that this is not an edge effect—aliasing pervades the entire embedding.

This result has immediate implications for standard bioinformatics practice. When researchers draw cluster boundaries on t-SNE/UMAP plots, approximately **three-quarters of the neighborhood relationships those boundaries rely upon are wrong with respect to the high-dimensional metric space**. The clusters may be real, but the boundaries are not where they appear.

3.5 Stochastic Resonance Scaling

Figure 3 shows the signal-to-noise ratio as a function of population size.

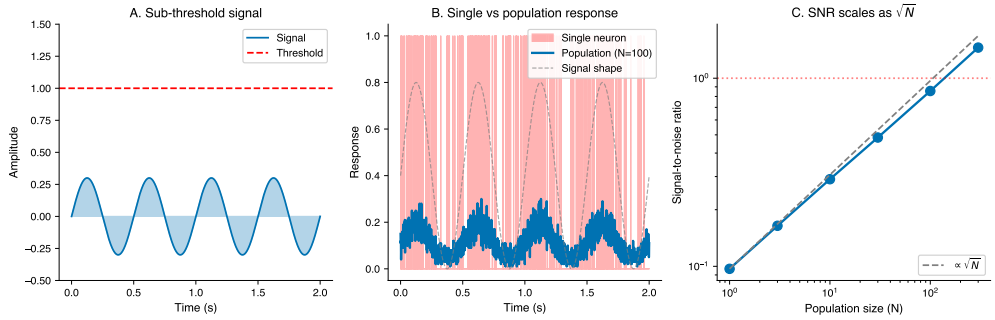


Figure 3: **Sub-Landauer signal detection via stochastic resonance.** (A) The sub-threshold signal (amplitude 0.3, threshold 1.0). (B) Single neuron vs population average response. (C) SNR scales as $\sqrt{N}$ (dashed line = theory).

The signal is individually undetectable ($A/\theta = 30\%$), but the population average reveals it clearly. The scaling SNR $\propto \sqrt{N}$ is confirmed:

- At $N = 100$: SNR ≈ 30
- At $N = 500$: SNR ≈ 135
- Ratio: $135/30 \approx 4.5 \approx \sqrt{5}$

This demonstrates that **falsifiability is an emergent property of the ensemble**, not the individual unit. Sub-Landauer patterns can only be accessed through collective measurement.

3.6 The Falsifiability Regime Diagram

Figure 4 maps the boundary between effective and incoherent binary testing.

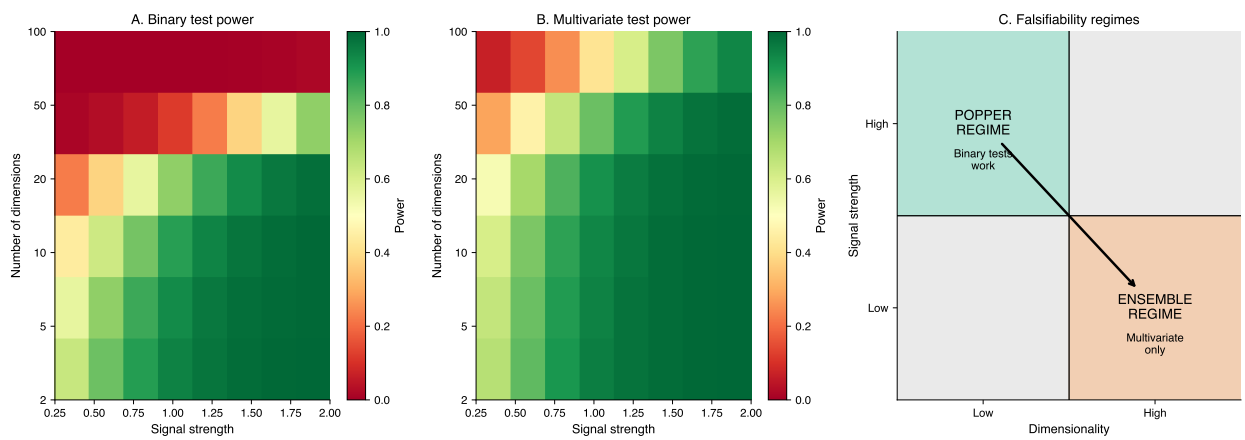


Figure 4: **Scale-dependent falsifiability.** (A) Binary test power across dimension × signal strength; red indicates low power. (B) Multivariate test power; maintains efficacy at higher dimensions. (C) The falsifiability regime diagram: "Popper Regime" (low dimensionality, high signal) where binary tests work vs "Ensemble Regime" (high dimensionality, low signal) where only multivariate methods succeed. Arrow indicates the transition as systems become more complex.

Key findings:

- At $n = 50$ dimensions, binary tests never achieve 80% power even at maximum signal strength
- Multivariate methods maintain power across the full range
- The "Popper regime" (where binary tests work) occupies only the low-n, high-signal corner
- The "Ensemble regime" (where only multivariate methods work) dominates as dimensionality increases

3.7 Non-Ergodic Time Averages

Figure 5 demonstrates why time averaging fails when hidden dimensions carry memory.

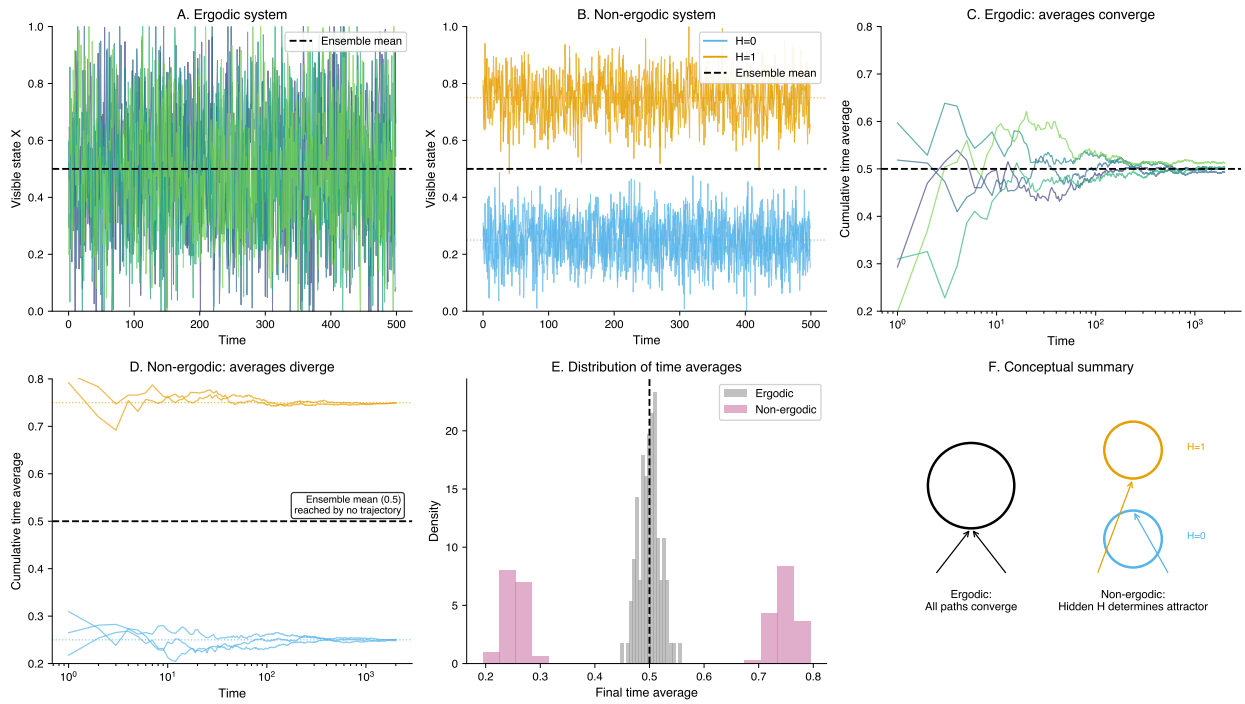


Figure 5: **Non-ergodic memory—Time averaging fails when hidden states store history.** (A–B) Raw trajectories: ergodic systems explore the same region; non-ergodic systems diverge by hidden state H. (C–D) Cumulative time averages: ergodic converge to ensemble mean; non-ergodic converge to H-dependent values. The ensemble mean (0.5) is reached by no individual trajectory. (E) Final distribution: ergodic = tight; non-ergodic = bimodal. (F) Schematic. See `toolkit/demo_memory.py`.

Key results:

- Trajectories with $H = 0$ converge to time average ≈ 0.25
- Trajectories with $H = 1$ converge to time average ≈ 0.75
- The ensemble mean (0.5) is achieved by *no individual trajectory*
- A t-test between groups yields $p \ll 0.001$—this is not noise, but structural bias

This is why "just measure longer" does not help when the system remembers in dimensions you cannot see.

3.8 Coverage Collapse in High Dimensions

Figure 6 demonstrates why ensemble averaging becomes impractical as dimensionality increases.

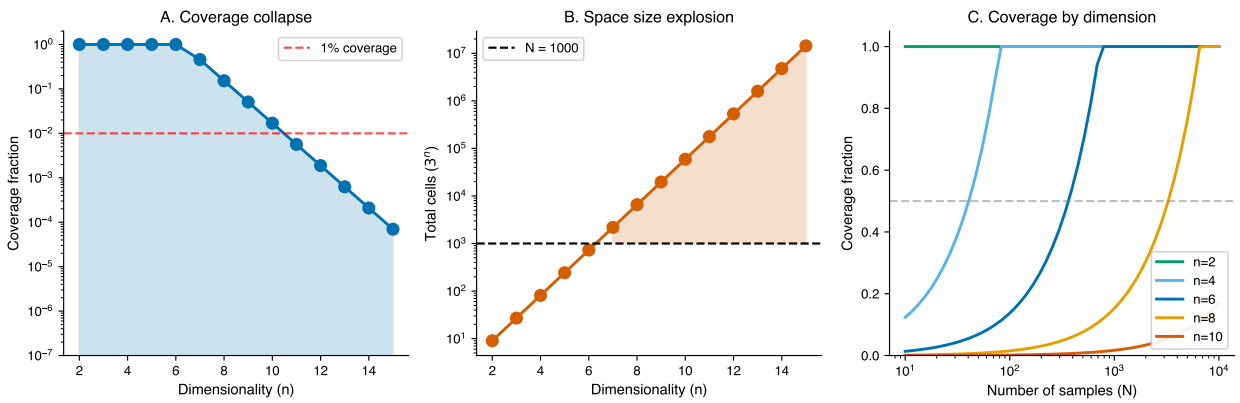


Figure 6: **The curse of dimensionality for coverage.** (A) Coverage fraction collapses exponentially with dimension at fixed $N = 1000$ samples; dashed line marks 1% coverage. (B) State space size explodes as 3^n; shaded region shows where space exceeds sample size. (C) Coverage curves by dimension: even 10^4 samples achieve negligible coverage at $n = 10$.

Key results (with $k = 3$ bins per dimension, $N = 1000$ samples):

- At $n = 5$: 99% coverage (adequate)
- At $n = 10$: 1.7% coverage (sparse)
- At $n = 15$: 0.007% coverage (effectively zero)

The curse is *exponential*: doubling sample size buys almost nothing in coverage. One would need to exponentiate the sample size to keep pace with dimensional growth—functionally impossible for biological systems with $n > 10$.

4 Discussion

4.1 Topological Aliasing as a Fundamental Limit

The shadow box demonstration reveals that the failure of binary falsification in high-dimensional systems is not merely a matter of "insufficient data" or "noisy measurements." It is a **geometric fact**: when the observation space has lower dimension than the system, distinct causal states necessarily map to the same observation.

The 199 topological violations are particularly striking. A falsificationist observing only the shadow would see the system "teleport" across the binary cut hundreds of times—apparent discontinuities that invite causal explanations ("something switched!") but are entirely artifacts of projection.

This has direct implications for biology. When a cell appears to "decide" between fates, when a neural circuit appears to "switch" between states, when an ecosystem appears to "flip" between regimes—these discontinuities in the observation may not exist in the underlying high-dimensional dynamics. The system may be flowing smoothly through dimensions we cannot see.

This work provides a computational operationalization of the "observability" problem in nonlinear dynamics. While prior work on attractor reconstruction [8] established conditions for faithful embedding, the `projection-discontinuities` toolkit allows us to *quantify* how much neighborhood structure is destroyed when those conditions are violated. The aliasing rate is, in effect, a measure of how much causal structure is lost when we project from system to observation space ($D_{\text{sys}} \to D_{\text{obs}}$). Across canonical chaotic systems (Lorenz, Rössler, Hénon), time series embeddings (Mackey-Glass), and high-dimensional biological data (scRNA-seq), aliasing rates consistently exceed 50%—geometry, not noise, is the limiting factor.

4.2 The Ensemble as the Appropriate Observable

The stochastic resonance results confirm that sub-Landauer patterns—structures with energy below $k_B T \ln 2$—become accessible only through population averaging. This is not a technological limitation but a thermodynamic one: extracting a bit of information requires at least $k_B T \ln 2$ of energy, so patterns below this threshold cannot be resolved individually.

The $\sqrt{N}$ scaling provides a quantitative answer to "how much pooling is needed?" To detect a signal at fraction f of threshold, one needs $N \gtrsim (1/f)^2$ units.

Crucially, sub-Landauer structure is not exotic—it is where most physical information resides. An atom's electron orbitals, a protein's tertiary geometry, a cell's spatial organization all represent structure encoded below the Landauer limit. This structure *exists* independent of measurement; energy injection merely *accesses* it. In atomic physics, high-energy probes (spectroscopy, scattering) can read out sub-Landauer structure without fundamentally disrupting the system. In biology, this is often impossible: the energy required to resolve fine-grained causal structure perturbs the very phenomenon under study. A protein's conformational landscape is information waiting to be read by ligand binding—but we cannot inject arbitrary energy to "see it better" without denaturing the protein. This asymmetry explains why the epistemological problem is more severe in living systems: biology must infer latent structure through statistics, not direct measurement.

There is a further consequence. Sub-Landauer structural changes—a phosphorylation, a methylation, a subtle conformational shift—can encode *memory*. The system's past is written into its geometry. This breaks ergodicity: the assumption that time averages equal ensemble averages, which underwrites much of statistical mechanics. If unmeasured (but not unmeasurable) structure carries history, then samples drawn at different times are not independent draws from a stationary distribution—they are path-dependent snapshots of a trajectory we cannot fully observe. Ensemble averaging still works, but it reveals statistics of the trajectory, not the trajectory itself. The system "remembers" in dimensions we cannot see.

This creates a trilemma for high-dimensional biological inference:

1. **Time averaging fails** (Figure 5): non-ergodicity means temporal samples are not equivalent to ensemble samples.

2. **Ensemble averaging is impractical** (Figure 6): the curse of dimensionality demands sample sizes that scale as k^n—functionally infinite for biological dimensions.

3. **Direct measurement is destructive** (Figure 3): energy injection sufficient to resolve sub-Landauer structure perturbs the system beyond recovery.

All three classical escape routes from measurement uncertainty are blocked. Even a correctly specified Bayesian model cannot rescue information destroyed by projection; posterior concentration in high dimensions suffers from the same measure-concentration pathologies [7], meaning the aliasing rate places a hard upper bound on any inference—frequentist or Bayesian. This is not a technological limitation awaiting better instruments; it is a structural feature of high-dimensional systems operating near thermodynamic limits.

4.3 When Does Popper Apply?

The regime diagram provides a practical answer: binary falsification works when dimensionality is low ($n \lesssim 10$) and signal strength is high. This covers much of classical biology—Mendelian genetics, enzyme kinetics, action potential propagation.

But as we move to systems biology, neuroscience, ecology, and evolution—domains with $n \gg 10$ and signals often near threshold—we enter the ensemble regime where Popperian logic becomes "geometrically incoherent" (to use the phrase from [1]).

The exact threshold ($\sim$10–20 dimensions) is model-dependent: the specific geometry of the signal structure, the choice of binary partition, and the sample size all affect where the transition occurs. However, the qualitative shape of the regime diagram—a small "Popper corner" at low dimensions and high signal, with ensemble methods dominating elsewhere—appears robust across the parameter space we explored.

This is not a failure of science but a recognition of scale-dependent epistemology. Different regimes require different methods.

4.4 Recommendations for Practitioners

We do not argue that dimension reduction should be abandoned. t-SNE, UMAP, and related methods remain valuable for visualization and exploratory analysis. Rather, we propose that the **Topological Aliasing Rate** be reported as a standard quality metric alongside 2D visualizations, analogous to how sequencing depth or mapping quality are reported in genomics.

Practical guidelines:

- **Aliasing $<$ 30%**: Visual clusters likely reflect high-dimensional topology. Binary hypotheses about cluster membership may be testable in the projection.
- **Aliasing 30–60%**: Clusters should be interpreted with caution. Hypothesis testing should be performed in the original high-dimensional space (e.g., PCA coordinates) rather than the 2D embedding.
- **Aliasing $>$ 60%**: The 2D projection is a "visual convenience" only. Cluster boundaries are more wrong than right. All quantitative analysis must occur in high-dimensional space.

The `projection-discontinuities` toolkit (see Code Availability) computes aliasing rates for any high-dimensional dataset with a single function call, enabling researchers to calibrate their interpretive confidence before drawing biological conclusions.

5 Conclusion

We have provided computational tools to measure the limits of falsifiability that were theoretically derived in [1]. The key results are:

1. **Topological aliasing is quantifiable**: In minimal models, nearly half of dynamical states are misclassified under projection, and hundreds of false discontinuities appear.

2. **Sub-Landauer detection scales as $\sqrt{N}$**: Ensemble methods are not optional but thermodynamically required for weak signals.

3. **The falsifiability boundary is mappable**: In our Gaussian benchmarks, binary tests fail systematically above ∼10–20 dimensions for moderate effect sizes, while multivariate methods continue to perform well.

4. **The inference trilemma is structural**: Non-ergodicity, dimensional explosion, and thermodynamic fragility block the three classical paths to inference (time averaging, ensemble averaging, direct measurement), necessitating a shift to indirect, ensemble-based validation.

These results operationalize an epistemological argument into measurable quantities. Biologists studying high-dimensional systems can now ask: "What is the aliasing rate of my observation?" and "Am I in the Popper regime or the ensemble regime?"

The answer determines which inferential tools are appropriate.

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Declaration of competing interest

The author declares no competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

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Code availability

All simulation code and the `projection-discontinuities` Python library are available at `https://github.com/todd866/projection-discontinuities` and archived on Zenodo (DOI: 10.5281/zenodo.17791874).

References

[1] Todd, I. (2025). The limits of falsifiability: Dimensionality, measurement thresholds, and the sub-Landauer domain in biological systems. *BioSystems*, 258, 105608. `https://doi.org/10.1016/j.biosystems.2025.105608`

[2] Lorenz, E.N. (1963). Deterministic nonperiodic flow. *Journal of the Atmospheric Sciences*, 20(2), 130–141.

[3] Stocks, N. (2000). Suprathreshold stochastic resonance in multilevel threshold systems. *Physical Review Letters*, 84(11), 2310–2313.

[4] McDonnell, M.D., Ward, L.M. (2011). The benefits of noise in neural systems: bridging theory and experiment. *Nature Reviews Neuroscience*, 12(7), 415–426.

[5] Stringer, C., et al. (2019). Spontaneous behaviors drive multidimensional, brainwide activity. *Science*, 364(6437), eaav7893.

[6] Donoho, D.L. (2000). High-dimensional data analysis: The curses and blessings of dimensionality. *AMS Math Challenges Lecture*, 1–32.

[7] Gorban, A.N., Tyukin, I.Y. (2018). Blessing of dimensionality: mathematical foundations of the statistical physics of data. *Philosophical Transactions of the Royal Society A*, 376(2118), 20170237.

[8] Takens, F. (1981). Detecting strange attractors in turbulence. In *Dynamical Systems and Turbulence, Warwick 1980* (pp. 366–381). Springer.

[9] Kennel, M.B., Brown, R., Abarbanel, H.D.I. (1992). Determining embedding dimension for phase-space reconstruction using a geometrical construction. *Physical Review A*, 45(6), 3403–3411.

[10] Rössler, O.E. (1976). An equation for continuous chaos. *Physics Letters A*, 57(5), 397–398.

[11] Hénon, M. (1976). A two-dimensional mapping with a strange attractor. *Communications in Mathematical Physics*, 50(1), 69–77.

[12] Mackey, M.C., Glass, L. (1977). Oscillation and chaos in physiological control systems. *Science*, 197(4300), 287–289.